

Skill Week-11

Session-21

Task 1: To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components.

```
nirvighna@DESKTOP-VS57D2I:~$ mkdir myproject
nirvighna@DESKTOP-VS57D2I:~$ cd myproject
nirvighna@DESKTOP-VS57D2I:~$ myproject$ nano calculator.h
nirvighna@DESKTOP-VS57D2I:~$ myproject$ nano calculator.c
nirvighna@DESKTOP-VS57D2I:~$ myproject$ nano main.c
nirvighna@DESKTOP-VS57D2I:~$ myproject$ gcc main.c calculator.c -o program
nirvighna@DESKTOP-VS57D2I:~$ myproject$ ./program
Addition = 25
Division = 4
nirvighna@DESKTOP-VS57D2I:~$ myproject$ |
```

```
GNU nano 8.7.1
#ifndef CALCULATOR_H
#define CALCULATOR_H

int add(int a, int b);
int divide(int a, int b);

#endif
```

```
GNU nano 8.7.1
#include "calculator.h"

int add(int a, int b)
{
    return a + b;
}

int divide(int a, int b)
{
    if (b == 0)
    {
        return -1;
    }

    return a / b;
}
```

```
GNU nano 8.7.1
#include <stdio.h>
#include "calculator.h"

int main()
{
    int a = 20;
    int b = 5;

    printf("Addition = %d\n", add(a, b));

    int result = divide(a, b);

    if (result == -1)
    {
        printf("Error: Cannot divide by zero\n");
    }
    else
    {
        printf("Division = %d\n", result);
    }

    return 0;
}
```

Task 2: To Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios.

```
nirvighna@DESKTOP-VS57D2I:~$ nano error.c
nirvighna@DESKTOP-VS57D2I:~$ gcc error.c -o error
nirvighna@DESKTOP-VS57D2I:~$ ./error
Floating point exception ./error
nirvighna@DESKTOP-VS57D2I:~$ ./program 2> error.log
Addition = 25
Division = 4
nirvighna@DESKTOP-VS57D2I:~$ cat error.log
nirvighna@DESKTOP-VS57D2I:~$
```

```
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int a = 10;
    int b = 0;

    printf("%d\n", a / b);

    return 0;
}
```

Session-22

Task 1: To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

```
nirvighna@DESKTOP-VS57D2I:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [526 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [674 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [123 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [177 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [56.3 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [437 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [86.2 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [159 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [302 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [96.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [190 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [457 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [90.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3172 B]
Get:22 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 3978 kB in 3s (1377 kB/s)
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
nirvighna@DESKTOP-VS57D2I:~$ sudo apt install valgrind
valgrind is already the newest version (1:3.26.0-ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 63
nirvighna@DESKTOP-VS57D2I:~$ valgrind --version
valgrind-3.26.0
```

```

nirvighna@DESKTOP-VS57D2I:~$ nano memory.c
nirvighna@DESKTOP-VS57D2I:~$ gcc -g memory.c -o memory
nirvighna@DESKTOP-VS57D2I:~$ valgrind --leak-check=full ./memory
==2107== Memcheck, a memory error detector
==2107== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2107== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2107== Command: ./memory
==2107==
10
==2107==
==2107== HEAP SUMMARY:
==2107==   in use at exit: 20 bytes in 1 blocks
==2107==   total heap usage: 2 allocs, 1 frees, 1,044 bytes allocated
==2107==
==2107== 20 bytes in 1 blocks are definitely lost in loss record 1 of 1
==2107==    at 0x4850858: malloc (vg_replace_malloc.c:447)
==2107==    by 0x400117E: main (memory.c:6)
==2107==
==2107== LEAK SUMMARY:
==2107==    definitely lost: 20 bytes in 1 blocks
==2107==    indirectly lost: 0 bytes in 0 blocks
==2107==    possibly lost: 0 bytes in 0 blocks
==2107==    still reachable: 0 bytes in 0 blocks
==2107==    suppressed: 0 bytes in 0 blocks
==2107==
==2107== For lists of detected and suppressed errors, rerun with: -s
==2107== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
nirvighna@DESKTOP-VS57D2I:~$

```

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *ptr = malloc(5 * sizeof(int));

    ptr[0] = 10;
    ptr[1] = 20;

    printf("%d\n", ptr[0]);

    return 0;
}

```

```
nirvighna@DESKTOP-VS57D2I:~$ nano memory.c
nirvighna@DESKTOP-VS57D2I:~$ gcc -g memory.c -o memory
nirvighna@DESKTOP-VS57D2I:~$ valgrind --leak-check=full ./memory
==2233== Memcheck, a memory error detector
==2233== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2233== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2233== Command: ./memory
==2233==
10
==2233==
==2233== HEAP SUMMARY:
==2233==    in use at exit: 0 bytes in 0 blocks
==2233==   total heap usage: 2 allocs, 2 frees, 1,044 bytes allocated
==2233==
==2233== All heap blocks were freed -- no leaks are possible
==2233==
==2233== For lists of detected and suppressed errors, rerun with: -s
==2233== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
nirvighna@DESKTOP-VS57D2I:~$
```

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *ptr = malloc(5 * sizeof(int));

    ptr[0] = 10;
    ptr[1] = 20;

    printf("%d\n", ptr[0]);

    free(ptr);

    return 0;
}
```

Task 2: To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.

```
nirvighna@DESKTOP-VS57D2I:~$ nano calculator.c
nirvighna@DESKTOP-VS57D2I:~$ gcc calculator.c -o calculator
nirvighna@DESKTOP-VS57D2I:~$ echo "10 20" | ./calculator
30
nirvighna@DESKTOP-VS57D2I:~$ echo "50 25" | ./calculator
75
nirvighna@DESKTOP-VS57D2I:~$ nano test_cases.txt
nirvighna@DESKTOP-VS57D2I:~$ nano test.sh
nirvighna@DESKTOP-VS57D2I:~$ ./test.sh
-bash: ./test.sh: Permission denied
nirvighna@DESKTOP-VS57D2I:~$ chmod +x test.sh
nirvighna@DESKTOP-VS57D2I:~$ ./test.sh
Running Test Cases...
Test 1: PASS
Test 2: PASS
nirvighna@DESKTOP-VS57D2I:~$ ./test.sh > test_report.txt
nirvighna@DESKTOP-VS57D2I:~$ cat test_report.txt
Running Test Cases...
Test 1: PASS
Test 2: PASS
```

```
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int a, b;

    scanf("%d %d", &a, &b);

    printf("%d\n", a + b);

    return 0;
}
```

```
GNU nano 8.7.1
10 20
50 25
100 200
5 7
```

```
GNU nano 8.7.1
#!/bin/bash

echo "Running Test Cases..."

result=$(echo "10 20" | ./calculator)

if [ "$result" = "30" ]; then
    echo "Test 1: PASS"
else
    echo "Test 1: FAIL"
fi

result=$(echo "50 25" | ./calculator)

if [ "$result" = "75" ]; then
    echo "Test 2: PASS"
else
    echo "Test 2: FAIL"
fi
```

